

A De-Poissonization Reduction for the UMD Case of Remark 12.10

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2026-06-16

Status

This is a conditional reduction for Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. It does not prove the broad UMD case. It isolates a single symmetric de-Poissonization estimate whose verification for all UMD spaces would settle the remaining discrete aggregate-domination bottleneck.

The lane context is now as follows. The broad Banach-space statement is false in c_0 . The Hilbert-valued case is solved for every finite p . The $L^q(S)$ -valued case is solved for all $1 < q < \infty$ and all $p \geq 1$. The remaining question is whether UMD geometry alone forces the same de-Poissonization phenomenon that Hilbert and L^q spaces enjoy.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source states that the general local-martingale case is open and reduces the missing issue to independent increments.

...and the ξ_n independent and η_n being independent. Then (12.10) follows from (12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The continuation gives the aggregate-domination formulation and says that even the symmetric case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

The Conditional Property

Let X be a Banach space. If $Z_1, \dots, Z_n$ are independent symmetric X -valued random variables, put

$$\mu_Z = \sum_{i=1}^n \mathcal{L}(Z_i)|_{X \setminus \{0\}}.$$

Let Π_{μ_Z} denote the compound-Poisson X -valued random variable whose intensity measure is μ_Z . Equivalently, if

$$Z_i = B_i U_i, \quad p_i = \mathbb{P}(Z_i \neq 0),$$

where B_i is Bernoulli p_i and U_i has the conditional law of Z_i given $Z_i \neq 0$, then

$$\Pi_{\mu_Z} = \sum_i \sum_{k=1}^{N_i} U_{i,k},$$

where N_i are independent Poisson variables with means p_i , and $(U_{i,k})$ are independent copies of U_i . Since the Z_i are symmetric, Π_{μ_Z} is symmetric and mean zero whenever the relevant first moment exists.

Definition 1 (Symmetric p de-Poissonization). *Let $1 \leq p < \infty$. Say that X has symmetric p de-Poissonization if there is a constant $D_{p,X}$ such that every finite independent symmetric family $Z_1, \dots, Z_n$ with finite p -moments satisfies*

$$(\mathbb{E} \|\Pi_{\mu_Z}\|^p)^{1/p} \leq D_{p,X} \left(\mathbb{E} \left\| \sum_i Z_i \right\|^p \right)^{1/p}.$$

Main Conditional Reduction

Theorem 1. *Fix $1 \leq p < \infty$. If X has symmetric p de-Poissonization, then the symmetric discrete aggregate-domination problem in Remark 12.10 holds in X : whenever (ξ_i) and (η_j) are finite independent symmetric X -valued families and*

$$\sum_i \mathbb{P}(\xi_i \in B) \leq \sum_j \mathbb{P}(\eta_j \in B), \quad B \in \mathcal{B}(X \setminus \{0\}),$$

one has

$$\left(\mathbb{E} \left\| \sum_i \xi_i \right\|^p \right)^{1/p} \leq 2D_{p,X} \left(\mathbb{E} \left\| \sum_j \eta_j \right\|^p \right)^{1/p}.$$

Idea of the Proof

Aggregate domination is awkward for Bernoulli sums because each original increment can occur at most once. It becomes natural after Poissonization: if $\mu_\xi \leq \mu_\eta$, then the compound-Poisson sum with intensity μ_η is the sum of an independent copy of the compound-Poisson sum with intensity μ_ξ and an independent symmetric remainder.

Thus two facts suffice. First, for symmetric families, the original Bernoulli sum is universally controlled by its Poissonized aggregate. Second, the assumed de-Poissonization property controls the Poissonized dominating family by the original dominating family. The first fact is where the threshold $1 - e^{-1} > 1/2$ reappears from the L^q endpoint packet; unlike the positive L^q proof, no lattice order is needed because symmetry and Jensen replace positivity.

Proof

Lemma 1 (Lower Poissonization for symmetric sums). *For every Banach space X , every $1 \leq p < \infty$, and every finite independent symmetric family $Z_1, \dots, Z_n$,*

$$\left(\mathbb{E} \left\| \sum_i Z_i \right\|^p \right)^{1/p} \leq 2 \left(\mathbb{E} \|\Pi_{\mu_Z}\|^p \right)^{1/p}.$$

Proof. Write $Z_i = B_i U_i$, where B_i is Bernoulli $p_i = \mathbb{P}(Z_i \neq 0)$, and U_i has the conditional symmetric law of Z_i given $Z_i \neq 0$. Let R_i be independent Bernoulli variables with common parameter

$$c = 1 - e^{-1} > 1/2.$$

Set $S = \sum_i B_i U_i$ and $S_R = \sum_i R_i B_i U_i$. Since

$$S = S_R + \sum_i (1 - R_i) B_i U_i,$$

the triangle inequality in $L^p(\Omega; X)$ gives

$$\|S\|_{L^p} \leq \|S_R\|_{L^p} + \left\| \sum_i (1 - R_i) B_i U_i \right\|_{L^p}.$$

Because $1 - c < c$, the second thinning can be coupled as a sub-thinning of the first. For symmetric independent summands, adding back an independent symmetric remainder can only increase the L^p -norm by Jensen's inequality. Hence

$$\left\| \sum_i (1 - R_i) B_i U_i \right\|_{L^p} \leq \|S_R\|_{L^p},$$

and therefore $\|S\|_{L^p} \leq 2 \|S_R\|_{L^p}$.

Now let N_i be independent Poisson variables with means p_i . Since

$$\mathbb{P}(N_i \geq 1) = 1 - e^{-p_i} \geq (1 - e^{-1})p_i = cp_i, \quad 0 \leq p_i \leq 1,$$

the first nonzero Poisson layer can be coupled as S_R plus an independent symmetric remainder. Jensen's inequality again gives

$$\|S_R\|_{L^p} \leq \left\| \sum_i \mathbf{1}_{\{N_i \geq 1\}} U_{i,1} \right\|_{L^p} \leq \|\Pi_{\mu_Z}\|_{L^p}.$$

Combining the two estimates proves the lemma. $\square$

Lemma 2 (Poisson monotonicity under symmetric aggregate domination). *Let $\mu \leq \nu$ be finite symmetric Borel measures on $X \setminus \{0\}$, and let Π_μ, Π_ν be the corresponding compound-Poisson sums. Then*

$$\|\Pi_\mu\|_{L^p} \leq \|\Pi_\nu\|_{L^p}.$$

Proof. By Poisson superposition, Π_ν has the same law as $\Pi_\mu + \Pi_{\nu-\mu}$, where the two terms are independent. The measure $\nu - \mu$ is symmetric, so $\Pi_{\nu-\mu}$ is mean zero whenever integrable. Conditioning on Π_μ and applying Jensen's inequality to the convex map $x \mapsto \|x\|^p$ yields the claim. The general finite p -moment case follows by the usual truncation at $\|x\| \leq R$ and monotone convergence. $\square$

Proof of Theorem 1. Let

$$\mu_\xi = \sum_i \mathcal{L}(\xi_i)|_{X \setminus \{0\}}, \quad \mu_\eta = \sum_j \mathcal{L}(\eta_j)|_{X \setminus \{0\}}.$$

The hypothesis is $\mu_\xi \leq \mu_\eta$. Lemma 1 gives

$$\left(\mathbb{E} \left\| \sum_i \xi_i \right\|^p \right)^{1/p} \leq 2 \|\Pi_{\mu_\xi}\|_{L^p}.$$

By Lemma 2,

$$\|\Pi_{\mu_\xi}\|_{L^p} \leq \|\Pi_{\mu_\eta}\|_{L^p}.$$

Finally, symmetric p de-Poissonization for the family (η_j) gives

$$\|\Pi_{\mu_\eta}\|_{L^p} \leq D_{p,X} \left(\mathbb{E} \left\| \sum_j \eta_j \right\|^p \right)^{1/p}.$$

Combining the three estimates proves the theorem. $\square$

Relation to the General UMD Case

Yaroslavl'tsev's Remark 12.10 says that, by a standard symmetrization trick [2, Lemma 6.3], the discrete problem may be reduced to the symmetric case. Therefore the broad UMD problem is reduced to the following geometric question.

Does every UMD Banach space have symmetric p de-Poissonization for every $1 \leq p < \infty$?

This formulation also explains the existing lane results. Hilbert spaces have the property by square-function identities, Rosenthal estimates, and the fractional-moment Laplace comparison used in the Hilbert packet. The spaces $L^q(S)$, $1 < q < \infty$, have it by the Dirksen–Yaroslavl'tsev Burkholder–Rosenthal formula for $p > 1$ and by the positive $L^{q/2}$ Poissonization argument at the endpoint $p = 1$. The c_0 counterexample is exactly a failure of de-Poissonization: the deterministic Rademacher sum $\sum_{k=1}^N \varepsilon_k e_k$ has c_0 -norm 1, while the Poissonized or with-replacement version has signed occupancy maximum of order $\log N / \log \log N$.

Literature and Search Notes

The bounded search for this push used the local run indexes for 1907.11588, Remark 12.10, aggregate domination, UMD, Poissonization, and de-Poissonization. Web and local source searches also checked Montgomery-Smith–Pruss [3], Hitczenko–Montgomery-Smith [4, 5], and Banach-space type/cotype context from Hytönen–McIntosh–Portal [6]. Montgomery-Smith–Pruss give the universal one-sided comparison from a heterogeneous independent family to an iid aggregate model, but they explicitly note that the converse tail inequality fails even for symmetric real variables. The present packet isolates the moment-level converse needed after imposing UMD geometry.

Limitations

This packet is conditional. It does not prove that every UMD space has symmetric p de-Poissonization. It also does not give a UMD counterexample. The next decisive step is either:

- prove symmetric de-Poissonization from UMD geometry, perhaps through K -convexity, finite cotype, or a random-coordinate thinning theorem for Rademacher sums; or
- construct a UMD space and a finite vector/law list for which the Poissonized aggregate has unbounded L^p -norm relative to the original independent symmetric sum.

References

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